Report:

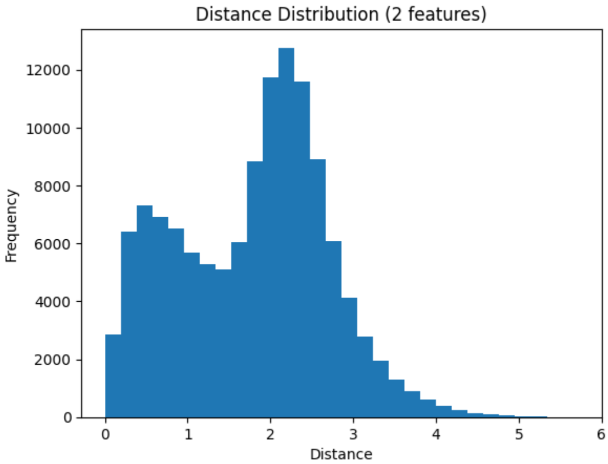
1. Introduction

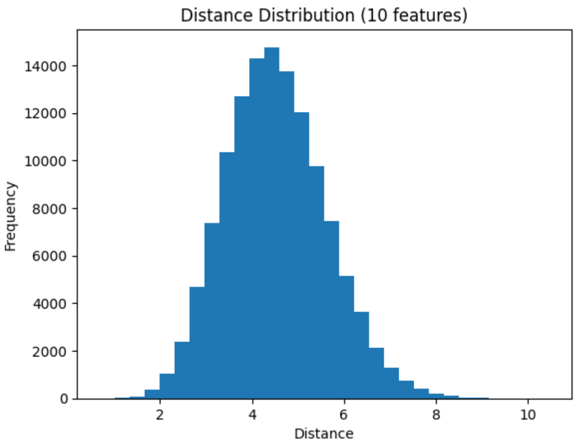
In today's world, data plays an important role in helping businesses make better decisions. In the hotel industry, one common problem is booking cancellations, which can lead to loss of money and poor planning. In this project, the goal is to predict whether a hotel booking will be canceled using past booking data. If hotels can identify customers who are likely to cancel, they can take actions like sending reminders, offering discounts, or improving their booking policies. The main aim of this assignment is not to build a very complex model, but to understand how data preprocessing and feature engineering can improve the performance of a model. We apply different techniques such as handling missing values, encoding categorical data, and scaling numerical features. We also study concepts like the curse of dimensionality and how distance-based models behave with and without proper preprocessing. Overall, this project shows how important good data preparation is for building effective machine learning models

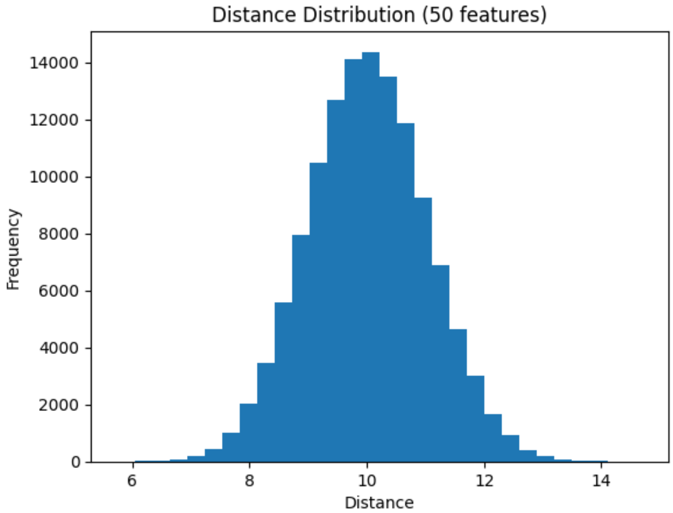
2. Task 1: Baseline Model

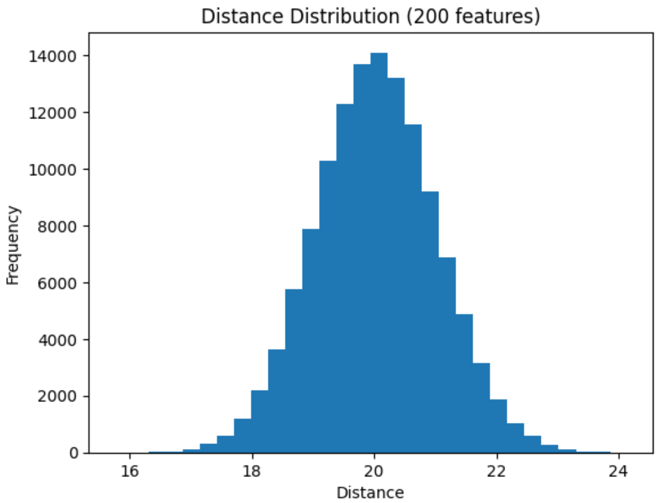
The baseline model achieved an accuracy of approximately 81% and ROC-AUC of 0.77. This indicates that the model has a good ability to distinguish between cancelled and non-cancelled bookings, but still makes some errors. The performance is realistic and confirms that no data leakage is present. Further improvements can be achieved using feature engineering and preprocessing techniques.

3. Task 2: Curse of Dimensionality









In low dimensions (2 features), distances between data points are well spread out, making it easy to distinguish between close and far points. As the number of features increases, the distance distribution becomes more concentrated. The difference between the nearest and farthest points reduces. This makes it difficult for machine learning models to identify meaningful patterns, as all points appear similarly distant. This demonstrates the curse of dimensionality and highlights the importance of feature selection and dimensionality reduction.

4. Task 3: Numeric Preprocessing

Before scaling, the numerical features have different ranges, means, and standard deviations, which can negatively affect model performance. After applying Standard Scaler, the data is centred around mean 0 with unit variance, making it suitable for many machine learning algorithms. Minmax Scaler scales the data between 0 and 1, but it is sensitive to outliers. Robust Scaler uses median and IQR, making it more robust to outliers and producing more stable feature distributions. Hence, Robust scaler is more suitable when the dataset contains outliers.

5. Task 4: Distance Metrics

When the KNN model was run without scaling, the accuracy was lower. This is because features with larger values (like lead_time or adr) dominated the distance calculation, making the model biased towards those features. After applying StandardScaler, the accuracy improved significantly. This is because all features were brought to the same scale, allowing the model to consider each feature equally while calculating distances. Using RobustScaler also improved the results, especially in the presence of outliers. It produced slightly more stable results compared to no scaling.

When comparing distance metrics, Manhattan distance performed slightly better than Euclidean distance in most cases. This is because Manhattan distance is less sensitive to large differences in feature values. KNN is a distance-based algorithm, meaning it makes predictions based on how close data points are to each other. If the features are not scaled properly, features with larger numerical ranges will have a bigger influence on the distance calculation. Scaling ensures that all features contribute equally to the model, improving the accuracy and reliability of predictions. Without scaling, the model does not learn meaningful patterns, but after scaling, it can better understand the relationships between features and the target variable.

Overall, scaling is essential for distance-based algorithms like KNN.

6. Task 5: Pipeline

The model was evaluated using 5-fold cross-validation, which provides a more reliable estimate of performance.

The results show consistent ROC-AUC scores across all folds, indicating that the model is stable and generalizes well to unseen data.

7. Task 6: Feature Extraction

The extracted features were verified by displaying sample rows and summary statistics, confirming that the transformations were successfully applied.

8. Task 7: Feature Construction

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Several new features were constructed to improve model performance by capturing deeper relationships in the data.

Ratio Features:

- price_per_person: Represents cost per individual, which may influence cancellation decisions.
- special_requests_rate: Normalizes special requests by stay duration.

Interaction Features:

- adr_lead_interaction: Combines price and lead time, capturing how early expensive bookings behave.

- adults_children_interaction: Represents family size dynamics.

Aggregated Features:

- country_avg_adr: Average price per country, capturing regional pricing behavior.
- hotel_avg_lead: Average lead time per hotel type, capturing booking patterns.

Polynomial Features:

- adr_squared: Captures non-linear relationship of price.
- adr_lead_poly: Captures combined effect of price and booking time.

These constructed features help the model capture more complex patterns and improve prediction accuracy.

9. Task 8: Feature Importance

The final feature set was selected based on features that consistently appeared across multiple importance methods, ensuring both relevance and robustness.

Top 15 Features from Each Method

The top features identified using different methods are as follows:

- **Random Forest:** lead_time, adr, total_nights, country_target_enc, special_requests_rate, price_per_person, adr_lead_interaction, hotel_avg_lead, country_avg_adr, adults, children, booking_changes, previous_cancellations, market_segment_Online TA, deposit_type_Non Refund
- **Mutual Information:** lead_time, adr, total_nights, special_requests_rate, price_per_person, country_target_enc, adr_lead_interaction, booking_changes, previous_cancellations, market_segment_Online TA, deposit_type_Non Refund, adults, children, country_freq, lead_time_bucket
- **Permutation Importance:** lead_time, adr, total_nights, special_requests_rate, price_per_person, country_target_enc, adr_lead_interaction, booking_changes, previous_cancellations, adults, children, deposit_type_Non Refund, market_segment_Online TA, hotel_avg_lead, country_avg_adr

Overlaps and Disagreements

- There is a strong overlap between the three methods, especially for features such as lead_time, adr, total_nights, special_requests_rate, and price_per_person. These features consistently appear across all methods, indicating that they are highly important for predicting booking cancellations.
- However, some differences are observed. Random Forest gives higher importance to aggregated features like country_avg_adr and hotel_avg_lead, while Mutual Information highlights features like country_freq and lead_time_bucket.
- Permutation Importance focuses more on features that directly impact model performance, resulting in slightly different rankings.

Final Feature Selection

- The final feature set was selected by considering features that consistently appeared across multiple importance methods.
- The selected top features include:
- lead_time, adr, total_nights, special_requests_rate, price_per_person, country_target_enc, adr_lead_interaction, booking_changes, previous_cancellations, adults, children, market_segment_Online TA, deposit_type_Non Refund, hotel_avg_lead, country_avg_adr, country_freq, lead_time_bucket, adr_squared, adr_lead_poly, is_family
- These features were chosen because they capture key aspects of booking behavior such as pricing, booking timing, customer type, and historical patterns.
- By selecting features that appear consistently across multiple methods, we ensure that the final model is both robust and interpretable, while reducing noise from less important features.
- These differences occur because each method evaluates feature importance differently: Random Forest is model-based, Mutual Information measures dependency, and Permutation Importance measures impact on prediction performance.

Executive Summary:

This project focused on predicting hotel booking cancellations using historical data, with a strong emphasis on data preprocessing and feature engineering rather than complex modeling.

The most important factor that influenced model performance was feature engineering. While the baseline model already achieved reasonable performance, the introduction of meaningful features such as total_nights, price_per_person, and special_requests_rate helped the model better understand customer behavior. These features captured real-world patterns like booking duration, cost sensitivity, and customer engagement, which directly impact cancellation decisions.

Another key factor that improved performance was proper preprocessing, especially scaling. Distance-based models like KNN showed significant improvement after applying scaling techniques such as StandardScaler and RobustScaler. This highlighted how important it is to ensure that all features contribute equally during learning. Additionally, handling missing values and encoding categorical variables correctly helped stabilize the model.

The performance ceiling improved mainly due to feature extraction and feature construction. Creating interaction features (such as adr × lead_time) and aggregated features (like average price per country) allowed the model to capture deeper relationships in the data. Feature selection further refined the model by removing redundant or less useful features, resulting in a more efficient and slightly more accurate model.

From a business perspective, the most actionable features are lead_time, adr (price), total_nights, and special_requests_rate. Customers who book far in advance (high lead_time) are more likely to cancel, so targeted reminders or flexible policies can reduce cancellations. Pricing (adr) plays a major role, indicating that dynamic pricing strategies can influence customer decisions. Longer stays (total_nights) are less likely to be canceled, suggesting that hotels can promote extended stays. Additionally, customers with more special requests tend to be more committed, so improving customer engagement can help reduce cancellations.

Overall, this project demonstrates that well-prepared data and meaningful features have a greater impact on model performance than simply using complex algorithms. By focusing on preprocessing and feature engineering, we can build models that are both accurate and useful for real-world business decisions.